



Intaliq 2.0 SRS Document

SE311: Software Requirements Analysis

Section: 1337

Version 1.0 approved

Instructor

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Revision History

Name	Date	Reason For Changes	Version
Yara Albugami	19/4/26	Initial draft	1.0 draft 1
Yara Albugami	20/4/26	Changes after inspection	1.0 Approved
Rose Al Rakan	21/4/26	Initial draft	1.0 draft 1
Rose Al Rakan	22/4/26	Changes after inspection	1.0 Approved
Wajd Albagma	23/4/2026	Initial draft	1.0 draft 1
Wajd Albagma	24/4/2026	Changes after inspection	1.0 Approved
Danah AlOmayrini	24/4/26	Initial draft	1.0 draft 1
Danah AlOmayrini	25/4/26	Changes after inspection	1.0 Approved

1.Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document presents the nonfunctional and functional requirements for release 1.0 for the Intaliq system.

The purpose of this document is intended to assist developers, testers, project managers, and stakeholders to clarify the system's requirements and anticipated behavior of the system. All of the requirements listed in the document are committed for release 1.0 and are regarded as high-priority, unless otherwise indicated.

This document is intended for project managers, testers, developers, stakeholders involved in the Intaliq system.

1.2 Document conventions

Standard software engineering requirements (SRS) conventions are followed in the document.

Requirements are labeled using unique identifiers with the prefix that identifies the type, such as OE (Operating Environment), CO (Constraint), AS (Assumption), and DE (Dependency), and FR (functional requirements) are used to label the requirements. Each requirement is assigned a unique identifier (ex: FR-1, FR-2) and identifiers are not reused even if the requirement is removed.

A hierarchical numbering system is used to organize and arrange requirements (ex: 3.x, 3.x.1, 3.x.2), which enhances clarity and traceability.

1.3 Project Scope

The Intaliq system is a fitness software for smart phones that was created to promote outdoor exercises in Saudi Arabia. Users are able to log activities such as running, cycling, and hiking, set personal fitness goals, and take part in fitness coach-led group sessions.

Through social connections and organized activities, the system seeks to increase the accessibility, motivation, and social engagement of outdoor fitness.

The Intaliq Vision and Scope documents provide a detailed explanation of the systems objectives, background, and business environment.

1.4 References

- Intaliq Vision and Scope Document
- IEEE Software Requirements Specification (SRS) Template
- Software Engineering Requirements course materials

2.Overall Description

2.1 Product perspective

The Intaliq system is a new standalone mobile application that is created to encourage outdoor fitness exercises. It presents a new platform that combines social interaction, activity, monitoring, and session management rather than replacing an existing system.

The system tracks user actions by interacting with mobile device components, like GPS and sensors. It may also integrate with external services like map services for location- based functionality .

Future releases of the system are anticipated to add fitness partners matching AI based suggestions.

2.2 User classes and characteristics

End Users (Participants)

End users are those who value outdoor fitness activities, such as cycling, hiking, and running. These user levels may range from beginners to experienced users.

End users may often utilize this system to explore sessions, track activities, and meet personal fitness objectives since they prefer an easy to use interface. While more experienced users could concentrate more on performance monitoring, beginners might need direction of encouragement

Fitness Coaches

Fitness coaches are in charge of planning, coordinating, and managing outdoor workout programs. Coaches specify the type of activity, time, location, and the degree of difficulty of the session.

They require system features that allow coaches to manage, create, monitor participant registration, manage session capacity, and ensure that the sessions are well organized and conducted safely.

Product Owners

Product owners are in charge of establishing system priorities and making sure they are in line with business goals. They focus on improving user engagement, system performance, and overall product value.

They mostly engage with the system indirectly through decision-making and monitoring.

2.3 Operating environment

OE-1: The system shall operate mobile devices with iOS and Android.

OE-2: The system shall require internet connection (Wi-Fi or mobile data) it's necessary for the system.

OE-3: The system shall use GPS/location services to enable location-based functionality such as displaying session locations.

2.4 Design and implementation constraints

CO-1: The system shall incorporate both Arabic and English languages.

CO-2: The system shall ensure the protection of the users data by following the appropriate privacy and security regulations.

CO-3: The system interface shall be designed to reduce user effort and accomplish essential functions by needing only a few steps to complete tasks such viewing sessions and booking.

CO-4: Time and budget constraints might limit the implementation of innovative features such as AI.

2.5 Assumptions and dependencies

AS-1: Users will use their smartphones with internet connectivity.

AS-2: Fitness coaches will be able to create and manage sessions within the system.

AS-3: Users are prepared to share basic data required for activity monitoring and system operation.

DE-1: The system may rely on external map services for location-based functionality.

DE-2: The system may rely on mobile device features such as GPS and sensors for accurate activity monitoring.

DE-3: AI services and other external integrations may be necessary for future functionalities.

3. System features

3.1 Book a session

Actors	Users
Description	A user can browse the available group sessions and book a session to take part in outdoor exercises, such as cycling, hiking, or running. The user can view available sessions, check session details, including the coach, and submit a request to join the session.
Data	User information. Session details: activity type, time, date, location, capacity of a session, intensity level of a session, and coach information.
Stimulus	User command on application selects and books a session
Response	Confirmation message that the booking request has been submitted. If accepted by the coach, the user is added to the session. If rejected, the user is notified.

Normal Scenario:

- 1- User opens the Intaliq app
- 2- User logs into the system
- 3- User browses available group sessions
- 4- User select session
- 5- User the review session details and coach details
- 6- User submits a booking request
- 7- The coach reviews the request
- 8- The coach accepts the request
- 9- The system confirms the booking and notifies the user

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message
- If the session is full —> the system prevents sending the request with a pop-up message
- If the coach rejects the request —> the system notifies the user
- If a system error occurs —> the request is not completed and the user is notified

3.2 Communicate with AI Chatbox

Actors	Users
Description	A user can chat with an AI Chatbox for fitness related advice.
Data	User entered-query
Stimulus	User command on AI chatbox chat
Response	AI Chatbox will answer user accordingly and suggest further help.

Normal Scenario:

- 1- User opens the Intaliq app
- 2- User logs into the system
- 3- User selects more tab
- 4- User select AI Chatbox
- 5- User select from given chat options or write their own
- 6- User submits the chat option by clicking
- 7- AI coach will answer the chosen query
- 8- User accepts answer and closes chat

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message

- If user asks for medical advice that should be only given from trained specialists
—> AI chatbox can refuse answer and end chat
- If a system error occurs —> the chat request is not completed and the user is notified

3.3 Enter activity goal

Actors	Users.
Description	A user can enter an activity goal they want to achieve.
Data	User information, activity type, specific goal to achieve.
Stimulus	User command on application.
Response	Confirmation by the system that the goal has been saved.

Normal scenario:

- 1- The user opens the Intaliq app.
- 2- The user logs into the system.
- 3- The user selects “Activity”.
- 5- The user selects “Activity Goal”.
- 4- The user enters their activity goal (activity type, goal).
- 5- The user submits the goal.
- 6- The system confirms the input.
- 7- The system saves the goal.
- 8- The system confirms the goal has been successfully created.

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message
- If any required field is missing —> The system prompts the user to fill in the missing information.
- If invalid data is entered —> The system displays an error message.
- If a system error occurs —> The goal is not saved.

3.4 Connect with partners

Actors	Users
Description	A user may find and connect with a fitness partner by browsing suggested partners based on location, goals, or similar activities. Users can send requests to partners.
Data	User information, other user (partner) information

Stimulus	User command on application.
Response	Confirmation by the system that the partner request has been saved. If request accepted, chat initiated If the request is denied, it will be clearly indicated near the profile of the requested partner.

Normal Scenario:

- 1- User opens the Intaliq app
- 2- User logs into the system
- 3- User selects more tab
- 4- User select Find Partners
- 5- User browse through profiles
- 6- User selects partner to connect with
- 7- System will confirm that request is sent
- 8- Partner accepts request
- 9- User and partner chat

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message
- If partner doesn't accept request —> no chat will open between user and partner
- If a system error occurs —> the chat request is not completed and the user is notified

3.6 Create group session

Actors	Fitness Coaches
Description	A coach can create a group session for users to join outdoor fitness activities by having details such as time, location, date, type of session, capacity, and difficulty level.
Data	Coach information, Session details: activity type, time, date, location, capacity of a session, intensity level of a session.
Stimulus	Coach command on application selects: "create session".
Response	Conformation by the system that the group session has been successfully created and is visible to users. If information is missing or invalid, the system notifies the coach to correct the input.

Normal scenario:

- 1- Coach opens the Intaliq app
- 2- Coach logs into the system
- 3- Coach selects “Create Group Session”
- 4- Coach enters the session details (activity type, date, time, location, capacity of users, intensity of session)
- 5- Coach submits the session
- 6- The system confirms the input
- 7- The system saves the session
- 8- The system confirms the session has been successfully created

Alternative Scenario:

- If the coach is not logged in —> system prompt login with a pop up message
- If any required field is missing —> the system prompts the coach to fill in the missing information
- If invalid data is entered —> the system display is an error message
- If a system error occurs —> The session is not saved and the coach is not notified.

3.7 Admit session members

Actors	Fitness Coaches.
Description	A coach can review users requests to join a group and decide whether to admit or reject them based on the session's requirements such as capacity, activity type, and difficulty level.
Data	User information, session details, membership request status, user fitness level, session capacity, session difficulty level.
Stimulus	Coach command on application selects: “admit session member”.
Response	Conformation by the system that the selected user has been admitted or rejected for the session. The system updates the session member list and notifies the user of the decision. If the user does not meet the requirements, the system informs the coach.

Normal scenario:

- 1- Coach opens the Intaliq app.
- 2- Coach logs into the system.
- 3- Coach selects the relevant session.
- 4- Coach opens the list of pending membership requests.
- 5- Coach reviews the user's request and profile information.
- 6- Coach selects “Accept” or “Reject”.

- 7- The system checks session capacity and session requirements.
- 8- The system updates the request status.
- 9- The system updates the session member list if the user is accepted.
- 10- The system sends a notification to the user with the decision.
- 11- The system confirms that the action has been completed successfully.

Alternative Scenario:

- If the coach is not logged in —> system prompt login with a pop up message
- If the session has reached maximum capacity —> the system prevents admission and informs the coach that no more users can be accepted.
- If the user's fitness level does not match the session difficulty level—> the system informs the coach and prevents admission.
- If no pending requests are available —> The system displays that there are no users waiting for approval.
- If a system error occurs —> The system does not save the decision and notifies the coach.

3.8 Session announcements

Actors	Fitness Coaches.
Description	A coach can create and publish announcements related to a group session in order to inform members about updates such as time changes, location changes, reminders, or special instructions.
Data	Coach information, session details, announcement content, announcement date and time, recipient session members.
Stimulus	Coach command on applications select: “session announcement”.
Response	Confirmation by the system that the announcement has been made successfully posted and delivered to the session members. If the content is missing or invalid, the system prompts the coach to correct it.

Normal scenario:

- 1- Coach opens the Intaliq app.
- 2- Coach logs into the system.
- 3- Coach selects the relevant group session.
- 4- Coach selects “Session Announcement”.
- 5- Coach writes the announcement content.
- 6- Coach submits the announcement.
- 7- The system validates the announcement content.
- 8- The system saves the announcement.

9- The system sends the announcement to all session members.

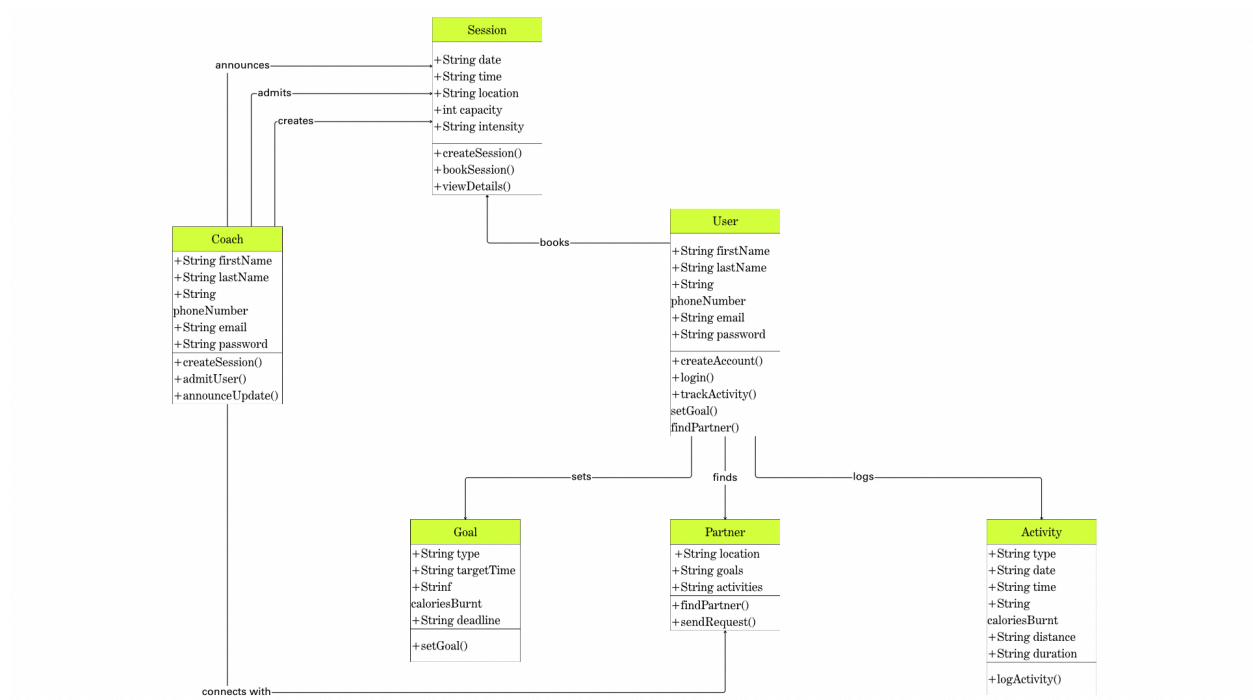
10- The system confirms that the announcement has been successfully posted.

Alternative Scenario:

- If the coach is not logged in —> system prompt login with a pop up message
- If the announcement field is empty—> the system prompts the coach to enter the missing content.
- If invalid data is entered—> the system displays an error message and asks the coach to correct it.
- If there are no registered members in the session —> The system saves the announcement but informs the coach that no users received it.
- If a system error occurs —> The announcement is not saved or sent, and the system notifies the coach.

4. Data requirements

4.1 Logical data model



4.2 Data dictionary

Session class type:

Data Name	Data type	Constraint
date	string	- By default, must follow the template: DD/MM/YYYY - Can't be an unrealistic date (far future/past)
time	string	- By default, must use the 24-hour clock time
location	string	- Must be located in Saudi Arabia - Must be permitted for group physical activities
capacity	int	- Must be more than 1 - Depending on event type and scale, location must accomodate
intensity	string	- Must fall under beginner, intermediate, or advanced

User class type:

Data Name	Data type	Constraint
firstName	string	- Must be less than 20 characters
lastName	string	- Must be less than 35 characters
phoneNumber	string	- Saudi Arabian phone number must follow the template: +966 5X-XXX-XXXX - Non-Saudi Arabian phone numbers must have their country code followed by their number

email	string	- Domain must be officially recognized - Email handle must already exist - Email handle should be less than 64 characters
password	string	- Must be 8 characters minimum with 1 upper case, 1 number, and 1 special character

Coach class type:

Data Name	Data type	Constraint
firstName	string	- Must be less than 20 characters
lastName	string	- Must be less than 20 characters
phoneNumber	string	- Saudi Arabian phone number must follow the template: +966 5X-XXX-XXXX - Non-Saudi Arabian phone numbers must have their country code followed by their number
email	string	- Domain must be officially recognized - Email handle must already exist - Email handle should be less than 64 characters
password	string	- Must be 8 characters minimum with 1 upper case, 1 number, and 1 special character

Goal class type:

Data Name	Data type	Constraint
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type	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running
targetTime	string	- Must follow template HH:MM
caloriesBurnt	string	- Must be more than 10 kCal
deadline	string	- By default, must follow the template: DD/MM/YYYY - Can't be an unrealistic date (past)

Partner class type:

Data Name	Data type	Constraint
location	string	- Must be located in Saudi Arabia - Must be permitted for group physical activities
goals	string	- Must relate to the 3 activities that Intaliq focuses on: Hiking, Biking, and Running - Other constraints mentioned in goal class must be noted
activities	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running

Activity class type:

Data Name	Data type	Constraint
type	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running
duration	string	- Must be 5 minutes and more - Shouldn't exceed a

		day
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4.3 Reports

No reports will be generated in Intaliq

4.4 Data acquisition, integrity, retention, and disposal

Data is acquired by user input:

- fitness goals
- activities
- personal information

Data acquired by coaches:

- session information
- personal information

Necessary techniques:

- backups
- checkpointing
- data accuracy verification

Data retention and disposal:

- After AI usage, prompts and input are deleted after the session.
- If output is significant, inputs and outputs are retained as formal record
- Intaliq system can differentiate between these types of deletion:
 - “active” deletion: User deletes specific information
 - ”passive” deletion: Automated policy. Usually after passive, information stays stored somewhere, and if not recovered after specific time duration, said information gets deleted permanently.

5. External interface requirements

5.1 User interfaces

UI-1: The Intaliq application shall provide a mobile user interface in both Arabic and English. Users shall switch languages from the settings page.

UI-2: All screens shall follow the same visual style, including:

- Green and white color palette.
- Rounded cards and buttons.
- A simple mobile layout optimized for one-handed use .
- Minimal text entry and large icons.

UI-3: The application shall use a bottom navigation bar to allow access to the main sections of the system: Home, Activities, Sessions, Goals, and More.

UI-4: The Home screen shall act as the main page from which users can navigate to session booking, activity logging, partner requests, and progress tracking.

UI-5: As shown in the prototype, all back and update buttons shall return the user to the previous main profile page after the action is completed.

UI-6: The shall provide separate screens for:

- Viewing and editing the profile.
- Booking and cancelling sessions.
- Logging activities.
- View goals and progress statistics.
- Using the AI Chatbox

UI-8: Each screen shall include clear confirmation and error messages near the related field.

UI-9: The application shall support touch-screen interaction in Android and IOS smartphones and tablets.

5.2 Software interfaces

SI-1: The Intaliq system shall connect to a cloud based database to store and retrieve user profiles, goals, activities, sessions, partner requests, and chat messages.

SI-2: The system shall integrate with an AI recommendation service to provide goal setting advice, recovery suggestions, progress analysis, and activity recommendations. The AI service shall not provide medical diagnosis treatment.

SI- 3: The system shall integrate with authentication services to support email login, password reset, and two factor authentication.

SI-4: The system shall integrate with push notification service to send session reminders, partner request update , coach announcements, and goal reminders.

SI-5: In future releases, the system may integrate with Google Maps API to display session locations and nearby activities.

5.3 Hardware interfaces

HI-1: The system may connect to wearable fitness devices such as smartwatches or fitness bands in future releases to receive:

- Heart rate.
- Distance covered.
- Calories burned.
- Activity duration.

5.4 Communications interfaces

CI-1: The Intaliq mobile application shall communicate with the backend server using HTTPS.

CI-2: The system shall support communication through both Wi-Fi and mobile data networks (4G/5G).

CI-3: The system shall send email messages to users for:

- Account verification during registration.
- Password reset requests.
- Two factor authentication codes.

CI-4: The system shall automatically synchronize user data activity logs, and partner requests whenever an internet connection is available.

CI-5: If the internet connection is lost, the application shall temporarily store unsent actions locally and automatically synchronize them when the connection is restored.

CI-6: The communication response time between the application and the server shall not exceed 5 seconds under normal operating conditions.

CI-7: The system shall prevent unauthorized access by requiring user authentication before transmitting personal information, session date, or activity records.

6. Quality attributes

6.1 Usability

USE-1. The system shall allow users to easily search for partners using filters with minimal effort.

USE-2. The system shall represent fitness statistics in a clear and understandable format.

6.2 Performance

PER-1. The system shall return results within 3 seconds.

PER-2. User statistics shall load within 4 seconds.

6.3 Security

SEC-1. The system shall protect personal data from unauthorized access.

SEC-2. The system shall implement strong password requirements (minimum length and special characters for complexity) for all user accounts.

6.4 Safety

SAF-1. The system shall provide workout recommendations customized to the user's health conditions and fitness level to avoid unsafe activities.

SAF-2. The system shall include warnings for potentially unsafe exercises.

6.x [others]

REL-1. The system shall be available at least 99% of the time.

AVA-1. The system shall handle multiple users simultaneously without failure.

7. Internationalization and localization requirements

Internationalization

INT-1. The system shall support multiple languages and character sets.

INT-2. The system shall support different cultural customs, including name formats and symbol usage.

Localization

LOC-1. The system shall allow users to select and switch between measurement systems.

LOC-2. The system shall support time zone differences based on user location.

8. Other requirements

Appendix A: Low Fidelity Prototype:

https://drive.google.com/file/d/1_Yti0aVDjRhgAfMks98Iyo4wRcyOKuvO/view?usp=drivesdk

Appendix B: Stakeholder Approval

Stakeholder: Sultanah AlOmayrini

Role: User type

Interview Date: Feb 14, 2026

Approval statement:

“I have reviewed the SRS document for Intaliq. I confirm that the features, specifically the ability to find fitness partners, join group sessions, track activities, and receive personalized recommendations, accurately reflect the needs and challenges I shared during the interview. The system’s focus on simplicity, clear interface design, and motivational features effectively supports user engagement in outdoor activities. I approve these requirements of the initial system release.”

Stakeholder signature: Sultanah AlOmayrini